Paper 2 assesses all content including content in **bold**.

This Edexcel International GCSE in Physics requires students to demonstrate an understanding of:

- forces and motion
- electricity
- waves
- energy resources and energy transfer
- solids, liquids and gases
- magnetism and electromagnetism
- radioactivity and particles.

Section 1: Forces and motion

a) Units

b) Movement and position

c) Forces, movement, shape and momentum

d) Astronomy

a) Units

Students will be assessed on their ability to:

1.1 use the following units: kilogram (kg), metre (m), metre/second (m/s), metre/second^2 (m/s^2), newton (N), second (s), newton per kilogram (N/kg), **kilogram metre/second (kg m/s)**.

b) Movement and position

Students will be assessed on their ability to:

1.2 plot and interpret distance-time graphs

1.3 know and use the relationship between average speed, distance moved and time:

$$\text{average speed} = \frac{\text{distance moved}}{\text{time taken}}$$

1.4 describe experiments to investigate the motion of everyday objects such as toy cars or tennis balls

1.5 know and use the relationship between acceleration, velocity and time:

$$\text{acceleration} = \frac{\text{change in velocity}}{\text{time taken}}$$

$$a = \frac{(v-u)}{t}$$

1.6 plot and interpret velocity-time graphs

1.7 determine acceleration from the gradient of a velocity-time graph

1.8 determine the distance travelled from the area between a velocity-time graph and the time axis.

c) Forces, movement, shape and momentum

Students will be assessed on their ability to:

1.9 describe the effects of forces between bodies such as changes in speed, shape or direction

1.10 identify different types of force such as gravitational or electrostatic

1.11 distinguish between vector and scalar quantities

1.12 understand that force is a vector quantity

1.13 find the resultant force of forces that act along a line

1.14 understand that friction is a force that opposes motion

1.15 know and use the relationship between unbalanced force, mass and acceleration:

force = mass × acceleration

$F = m \times a$

1.16 know and use the relationship between weight, mass and g:

weight = mass × g

$W = m \times g$

1.17 describe the forces acting on falling objects and explain why falling objects reach a terminal velocity

1.18 describe experiments to investigate the forces acting on falling objects, such as sycamore seeds or parachutes

1.19 describe the factors affecting vehicle stopping distance including speed, mass, road condition and reaction time

1.20 know and use the relationship between momentum, mass and velocity:

momentum = mass × velocity

$\boldsymbol{p = m \times v}$

1.21 use the idea of momentum to explain safety features

1.22 use the conservation of momentum to calculate the mass, velocity or momentum of objects

1.23 use the relationship between force, change in momentum and time taken:

$$\textbf{force} = \frac{\textbf{change in momentum}}{\textbf{time taken}}$$